

STEAM-3D

Sustainable Training in Epidemiological Analysis and Modelling for Data-Driven Decision Making in Africa

SOUTH AFRICA | MALAWI | MOZAMBIQUE | CAMEROON | BELGIUM

Global Health EDCTP3

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The project is supported by Global Health EDCTP3 and its members

Introduction: Principles and practice of evidence informed decision-making

Primer for STEAM3D delegates

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Cochrane Africa

Cochrane Africa

Cochrane Africa Network

Cape Town
16 October 2026

Trusted evidence.
Informed decisions.
Better health.

Cochrane Africa

Cochrane Africa is a network that aims to increase the use of evidence to inform healthcare decision-making in sub-Saharan Africa



http://africa.cochrane.org/

Cochrane Africa over time



Cochrane Africa

Cochrane Africa launch

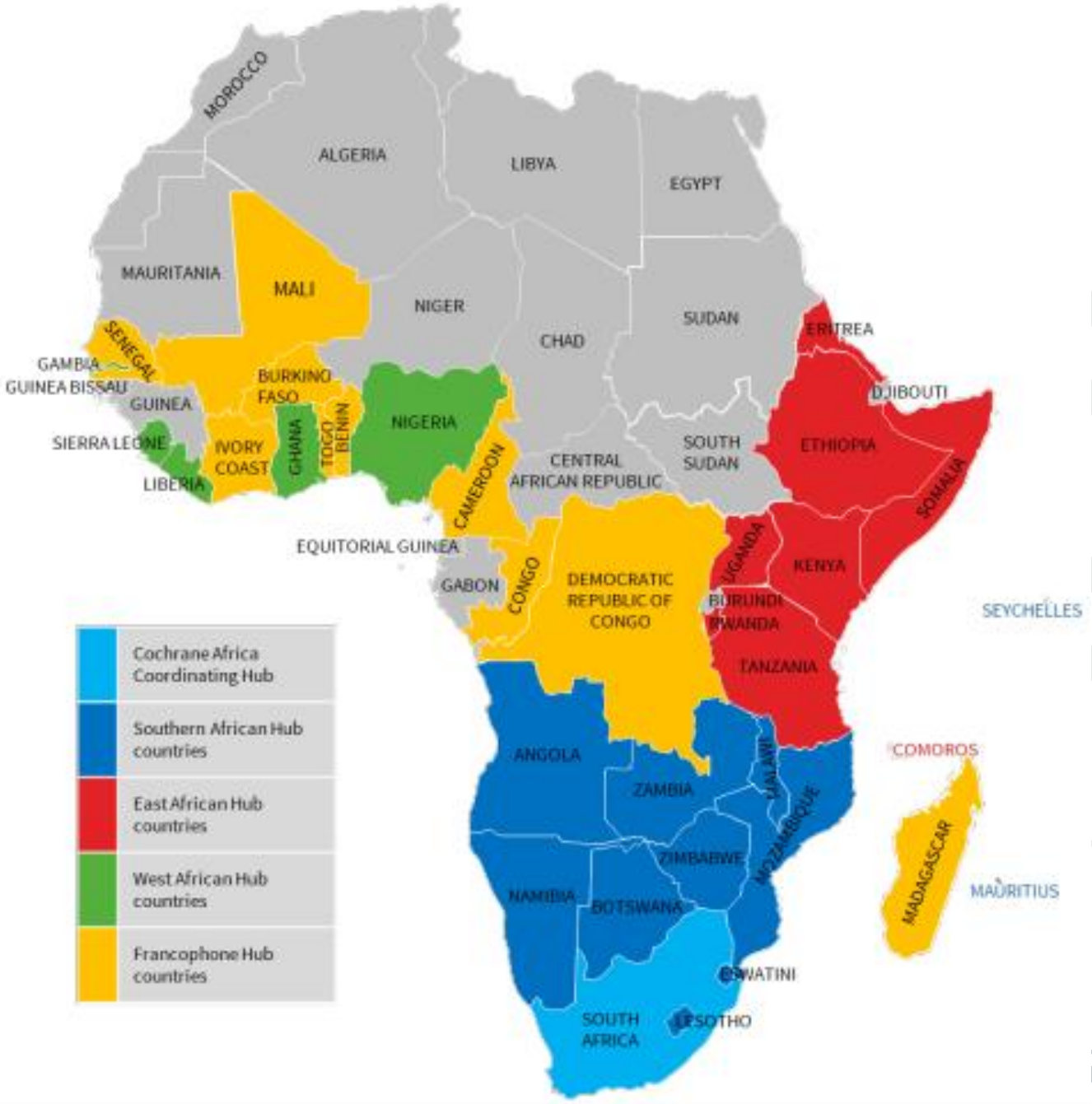


Global Evidence Synthesis Summit
Sep 2017
Cape Town, South Africa

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Cochrane Africa

Cochrane Africa Hubs



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Cochrane Africa

Guiding principles

1. Each country is unique
2. Prioritise to ensure relevance
3. Collaboration
4. Avoid unnecessary duplication
5. One step at a time
6. Build sustainability
7. Equity-focused in everything we do
8. Encourage diversity



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‘Knowledge to Action’ gaps

There are many examples of underuse and overuse of evidence in decision making and clinical care

- 1/3 patients do not get treatments of proven effectiveness
- 1/4 patients get care that is not needed or potentially harmful
- Up to 3/4 of patients don’t get the information they need for decision making
- Up to 1/2 of physicians don’t get the evidence they need for decision making
- Policy makers often don’t use evidence from systematic reviews to inform policies, despite its availability

http://ktclearinghouse.ca/knowledgebase/ktinhealthcare/1/1

Learning objectives

- Define evidence informed decision making (EIDM) and guidelines
- Brainstorm and share evidence on barriers to and enablers of EIDM
- Explain how EIDM is operationalised using multi-criteria approach and deliberation
- Discuss where modelling evidence fits into the Evidence to Decision framework

What is evidence?

Evidence alone does not make decisions.

Discussions of evidence-based practice and evidence-informed policymaking can generate debate about what exactly is **‘evidence’.**

Oxman AD, Lavis JN, Lewin S, Fretheim A. SUPPORT Tools for evidence-informed health Policymaking (STP) 1: What is evidence-informed policymaking? Health Research Policy and Systems. 2009;7(1):S1

Factual knowledge gained through observation or experimentation in support of a conclusion



Tacit evidence

Informal: opinions, values, habits



Scientific evidence

Explicit, systematic, replicable

Source: WHO. Evidence, policy, impact: WHO guide for evidence-informed decision-making. Geneva: WHO; 2021.

Two types of evidence, used together

Tacit

- Views of policymakers, clinicians, patients, citizens
- Dialogues, reports, policy documents
- Expert opinion: appraise the facts behind it



Scientific

- Primary: individual studies
- Secondary: systematic reviews
- Tertiary: guidelines, policy briefs

Complementary, not competing

Source: WHO. Evidence, policy, impact: WHO guide for evidence-informed decision-making. Geneva: WHO; 2021.

What influences decisions?



Political complexities



Party preferences



Other evidence



Social, religious and cultural norms (acceptability)



Relative benefit/harms of other options – cost, impact, equity, speed



Ecological impacts



Other actors – lobbyist, activists, media, epistemic communities,



Operational practicality

Evidence informed decision-making defined

“Evidence-informed decision-making (EIDM) entails identifying, appraising, and mobilizing the best available evidence for safe and effective health policy and programmes.”

In South Africa, we have an evolving decision- system, we state:

“EIDM is defined as a systematic and transparent approach that uses structured, replicable methods to identify, appraise, and apply evidence in decision-making processes, including implementation.”

World Health Organization. (2022). *Evidence, policy, impact: WHO guide for evidence-informed decision-making*. Geneva: World Health Organization. ISBN 978-92-4-003987-2.
South African National Department of Health. 2025. *Evidence-Informed Decision-Making Manual for the National Standard Treatment Guidelines and the Essential Medicine List: Applying the National Standards for Evidence-Informed Decision-Making to Develop Trustworthy Guidance Products* [Draft]. Pretoria, South Africa

Evidence informs, rather than dictates, decisions

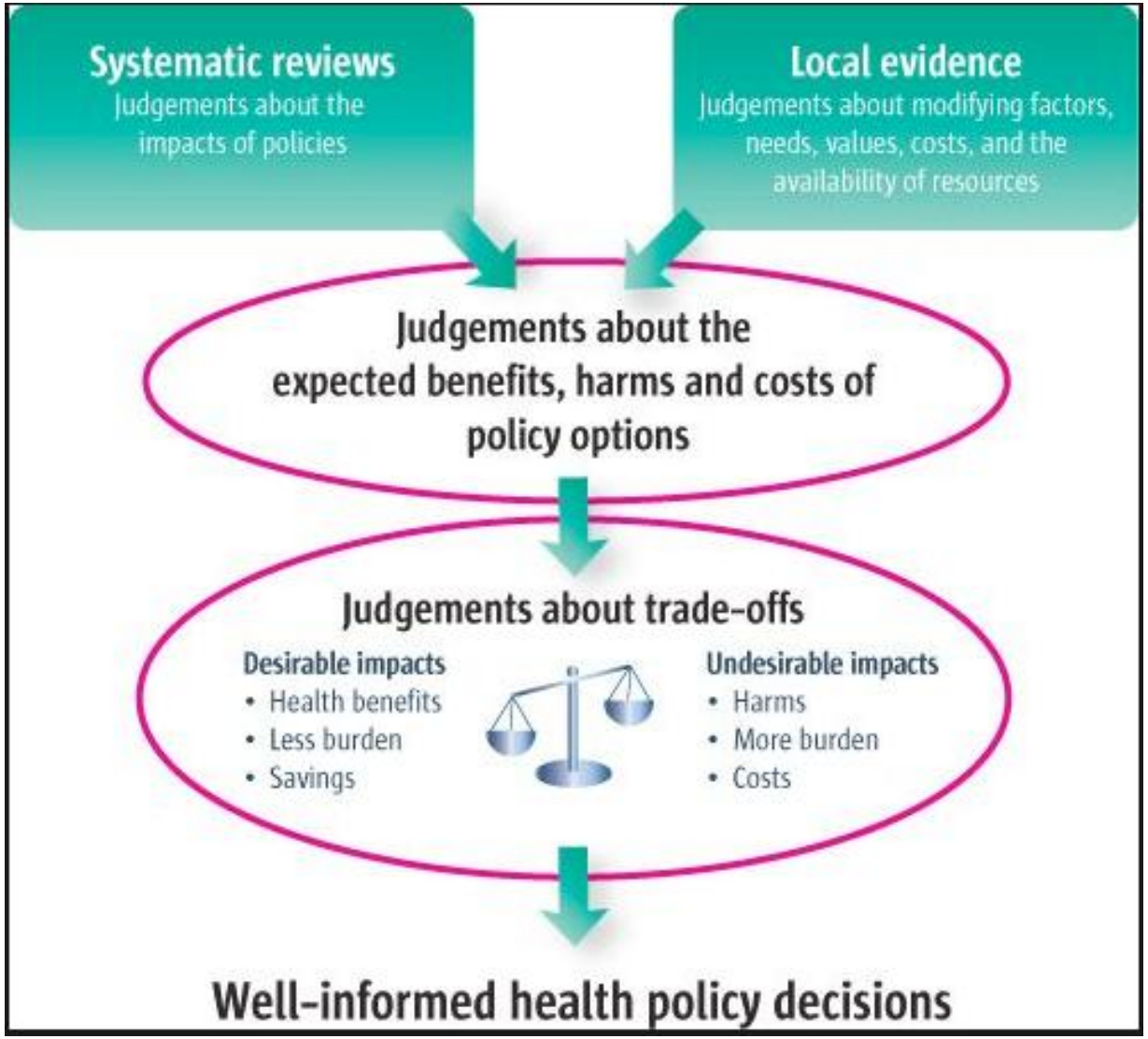
Research evidence is critical, but not the ONLY determinant of decisions

- **what else plays a role in decision-making?**

Evidence is only one factor



An example of the role of evidence in health policymaking



Oxman, A.D., Lavis, J.N., Lewin, S. *et al.* SUPPORT Tools for evidence-informed health Policymaking (STP) 1: What is evidence-informed policymaking?. *Health Res Policy Sys* **7** (Suppl 1), S1 (2009). https://doi.org/10.1186/1478-4505-7-S1-S1

Why isn’t research evidence always used for decision-making in health (and all sectors really)?

Barriers to and enablers of research uptake for decision-making for health?

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Why isn't research evidence always used for decision-making in health (and all sectors really)?

Systematic review – included studies about factors affecting the use of evidence in policy.

Studies: n = 145 in 59 countries

Oliver *et al. BMC Health Services Research* 2014, **14**:2
http://www.biomedcentral.com/1472-6963/14/2

BMC Health Services Research

RESEARCH ARTICLE Open Access

A systematic review of barriers to and facilitators of the use of evidence by policymakers

Kathryn Oliver[1*], Simon Innvar[2], Theo Lorenc[3], Jenny Woodman[4] and James Thomas[5]

"The most frequently reported barriers to evidence uptake were poor access to good quality relevant research, and lack of timely research output.
The most frequently reported facilitators were collaboration between researchers and policymakers, and improved relationships and skills.
There is an increasing amount of research into new models of knowledge transfer, and evaluations of interventions such as knowledge brokerage."

Oliver, K., Innvar, S., Lorenc, T. *et al.* A systematic review of barriers to and facilitators of the use of evidence by policymakers. *BMC Health Serv Res* **14**, 2 (2014). https://doi.org/10.1186/1472-6963-14-2

Experience with policymaking from STEAM3D delegates?

How can EIDM be operationalised for decisions?

How can we make good evidence informed decisions?

Guidelines – a tool for bridging evidence and practice

Guidelines are recommendations intended to **assist** providers and recipients of health care and other stakeholders to make **informed decisions**.

"….systematically developed evidence-based statements which assist providers, recipients and other stakeholders to make informed decisions about appropriate health interventions. Health interventions are defined broadly to include not only clinical procedures but also public health actions."

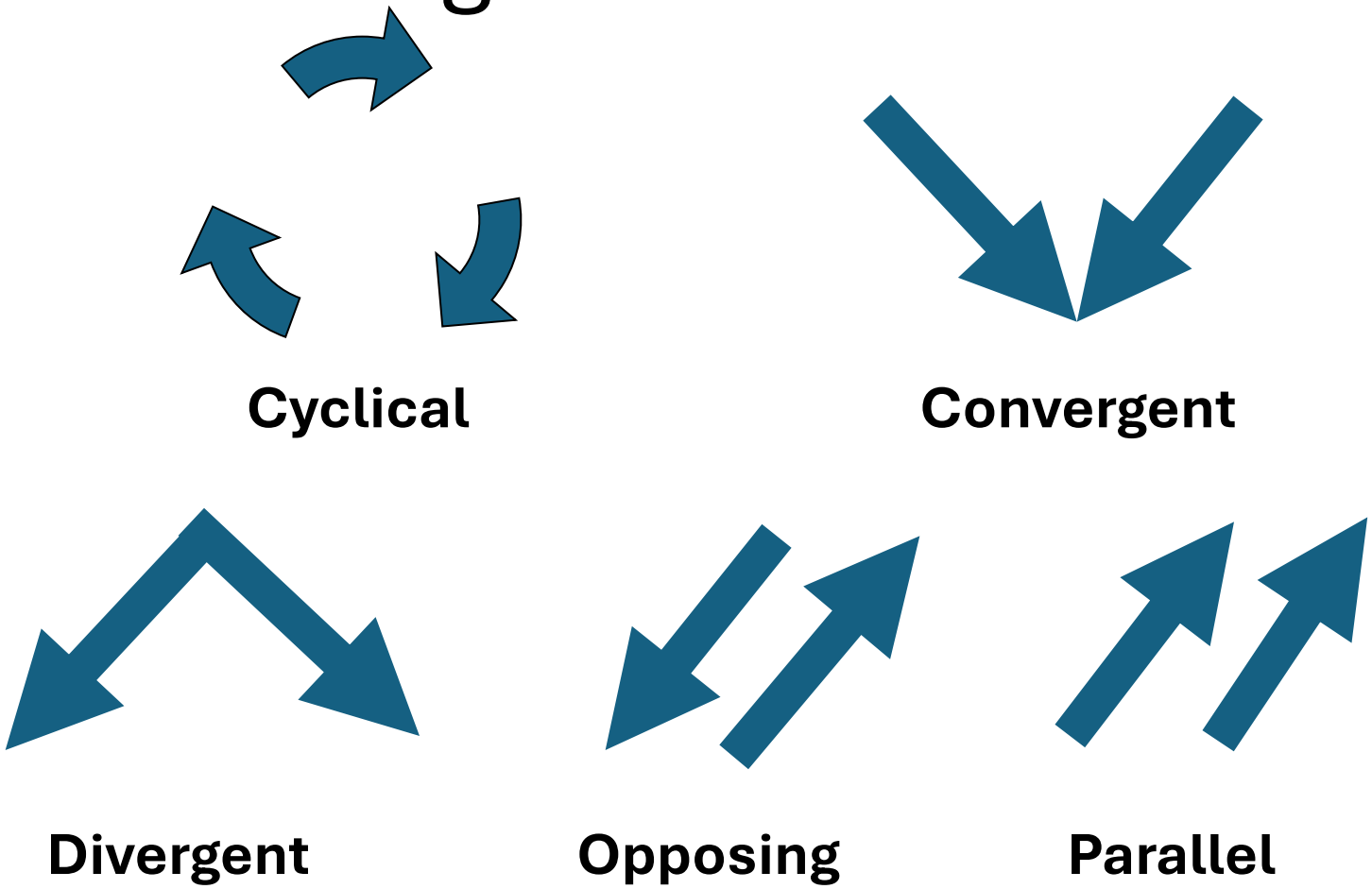
WHO 2003, 2007

GOBSAT guidelines

Traditional guidelines



Decision-making is not linear





RATING QUALITY OF EVIDENCE AND STRENGTH OF RECOMMENDATIONS

GRADE: an emerging consensus on rating quality of evidence and strength of recommendations

Guidelines are inconsistent in how they rate the quality of evidence and the strength of recommendations. This article explores the advantages of the GRADE system, which is increasingly being adopted by organisations worldwide

Approach

g of **R**ecommendations **A**ssessment,
ent and **E**valuation approach = a way of
ding how confident we are in our results

tic, transparent way of rating the
of evidence in systematic reviews (SRs)
practice guidelines (CPGs)

ng and determining the strength of
actice recommendations

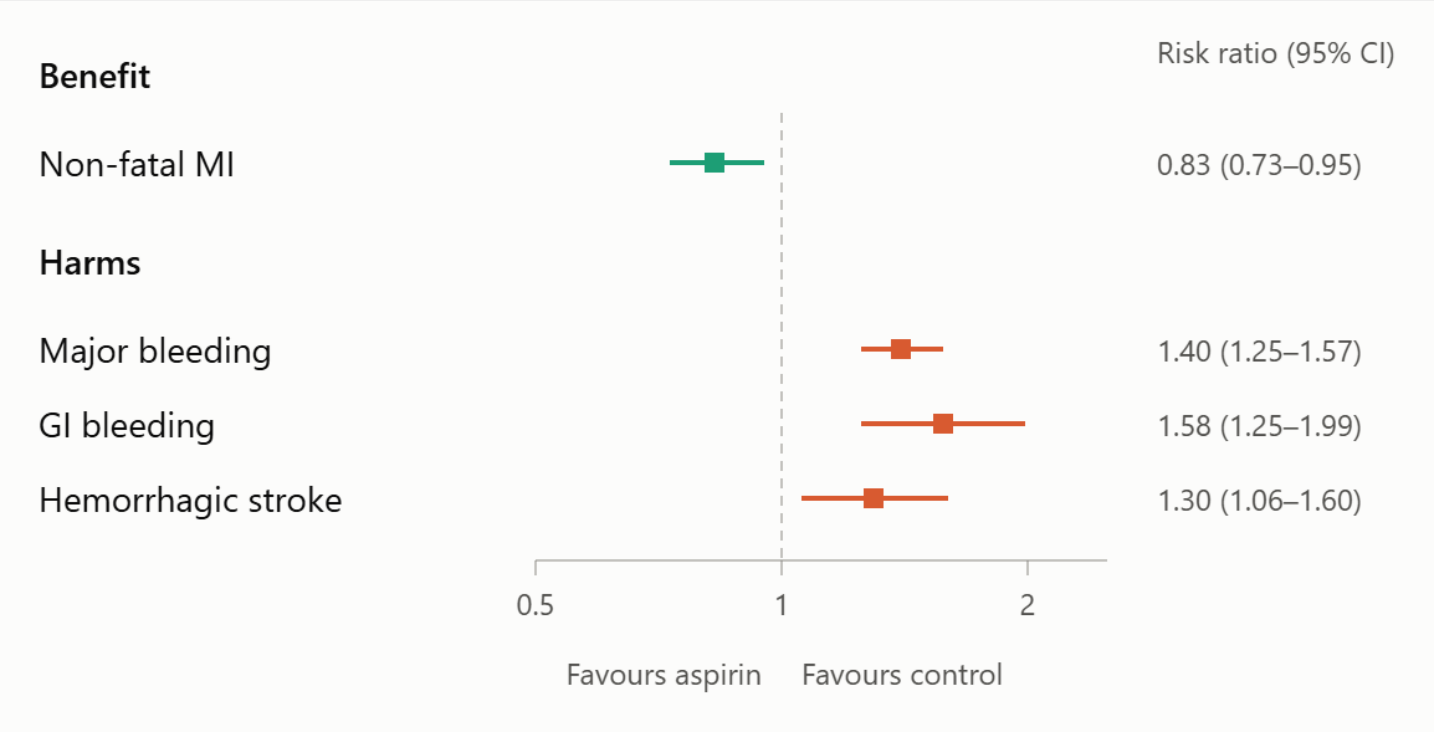
orking group

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EVIDENCE-TO-DECISION COLLABORATION

South Africa GRADE Network

WE USED TO ONLY LOOK AT BENEFITS & HARMS



THIS IS IMPORTANT BUT IS IT ENOUGH TO MAKE A DECISION?



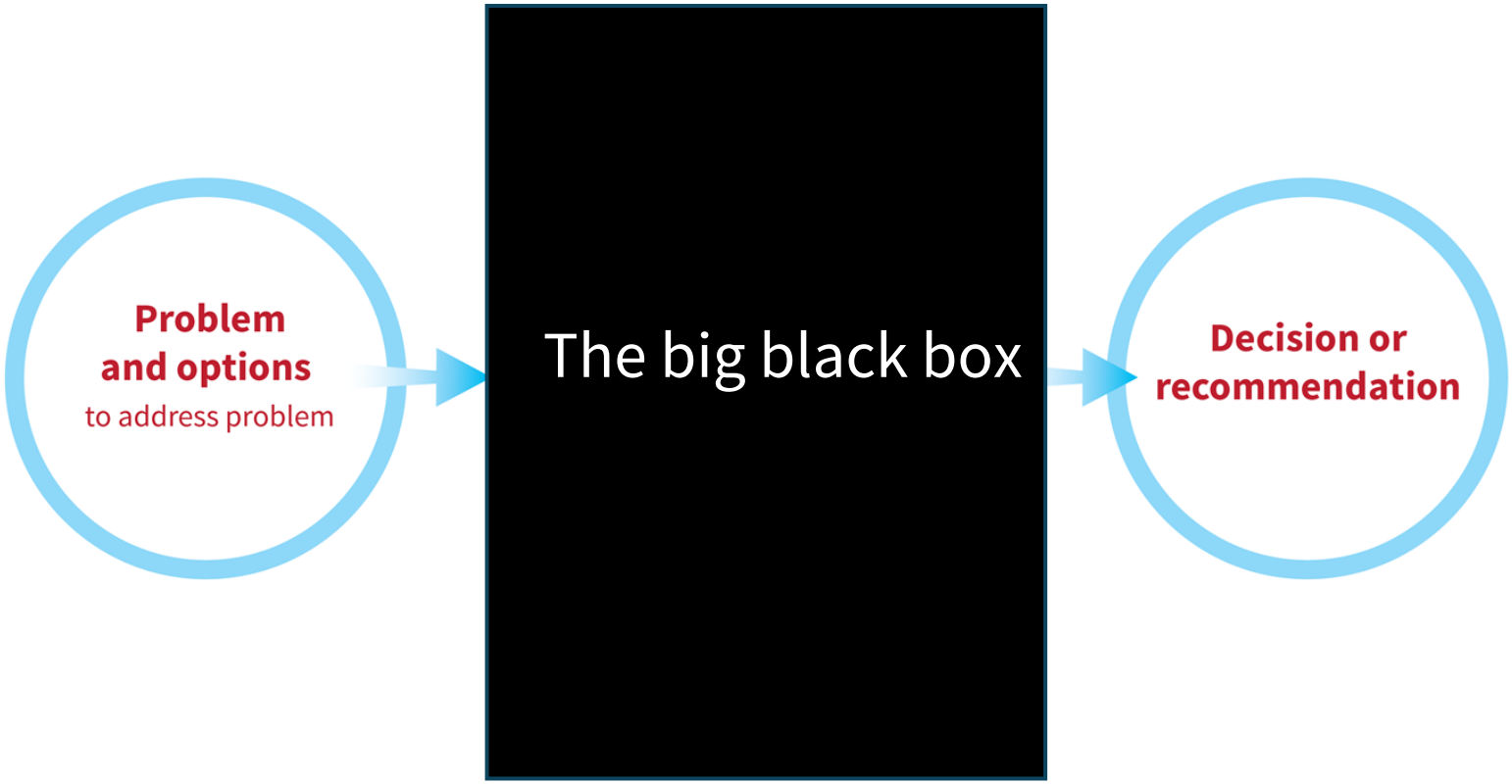
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EVIDENCE-TO-DECISION COLLABORATION

South Africa GRADE Network

GRADE evidence-to-decision frameworks

A move away from ***this*** type of decision making…





To **this** type of decision-making

GRADE Evidence to Decision Framework

Purpose

- Facilitate **structured**, **transparent** and **evidence-informed** decision-making across a wide range of health contexts
- Provide a common approach to formulating recommendations/decisions by helping panels/decision-makers **consider all key criteria systematically**

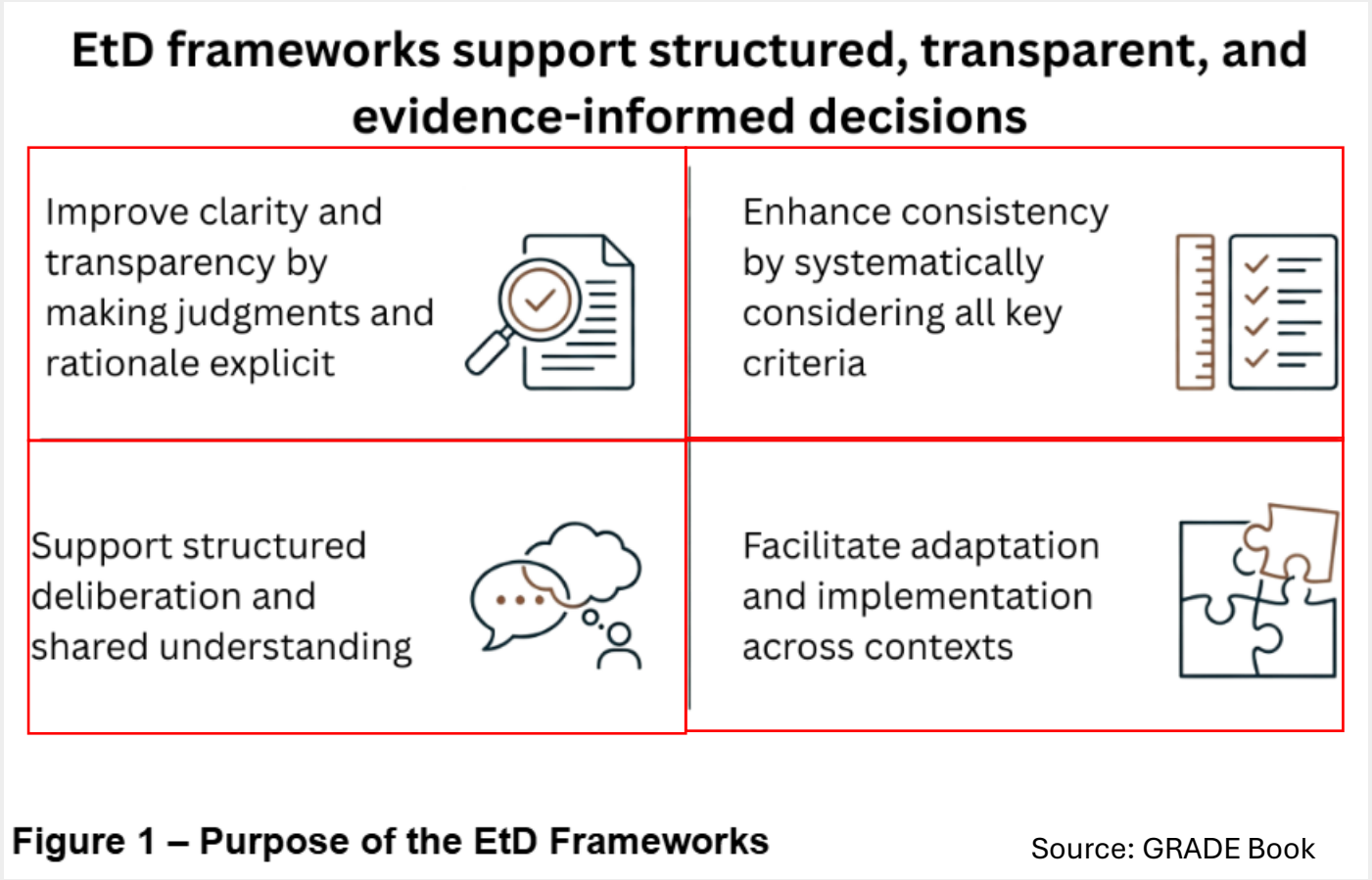


Figure 1 – Purpose of the EtD Frameworks

Source: GRADE Book



FRAMEWORK STRUCTURE

1 Formulate the question

2 Assessing the evidence

3 Drawing conclusions

Defining the PICO & the perspective

Applying criteria & making judgements

Recommendation & Justification



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Key evidence needed for decision-making

- Balance of benefits and harms
- Values
- Costs/resources and cost-effectiveness
- Acceptability
- Equity issues
- Feasibility

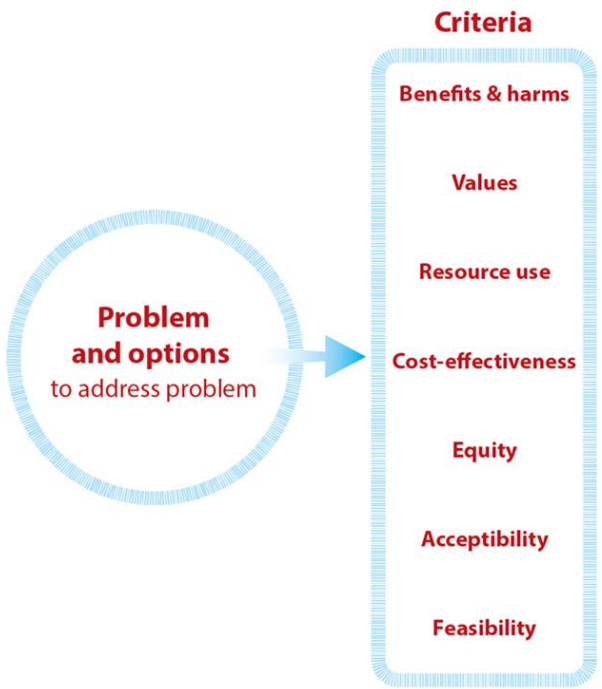


KEY STEPS FOR DECISION-MAKING

- Clear guideline question
- Prioritisation of outcomes
- Assessment of evidence
- Recommendation / Decision

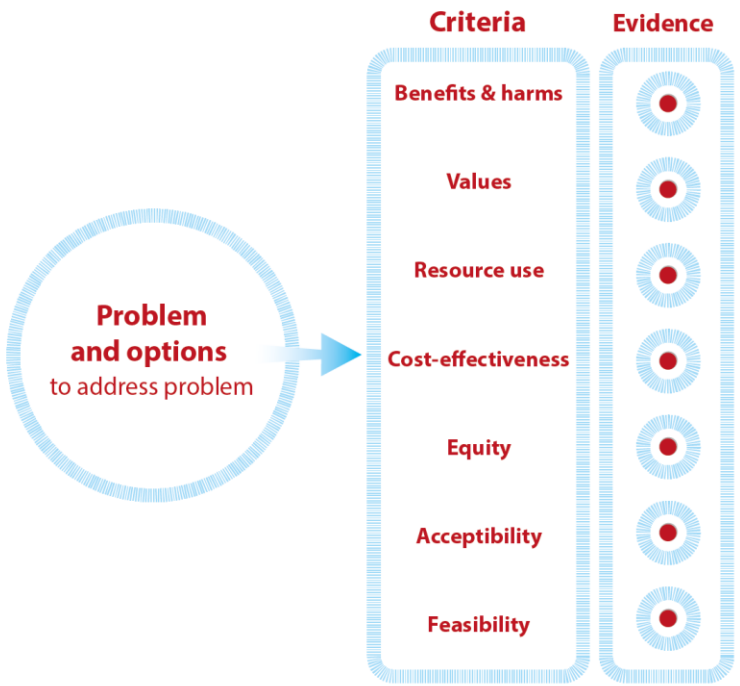


Criteria for considering a decision



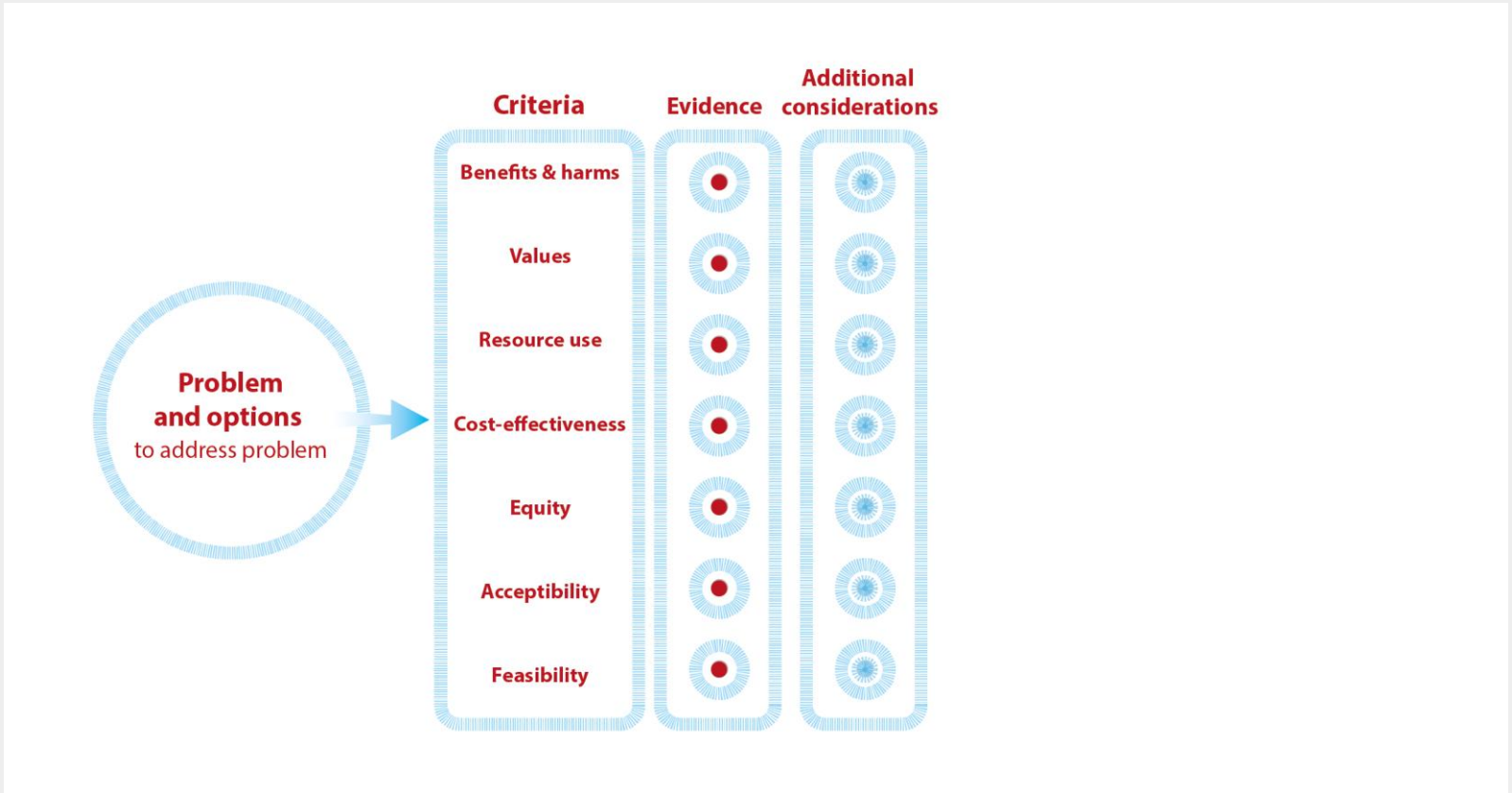


Ensuring transparency when evidence is lacking and presenting the best available evidence





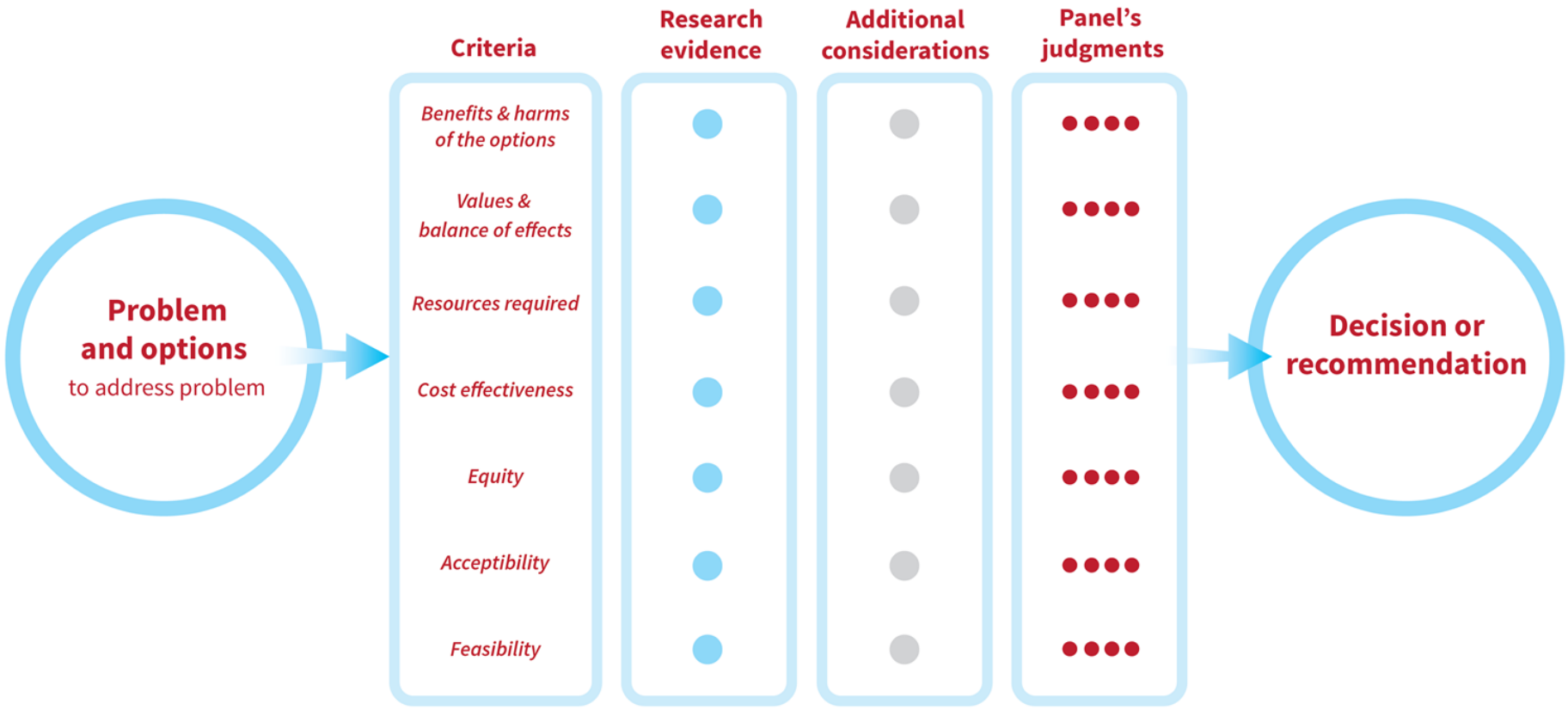
Providing space for other important information



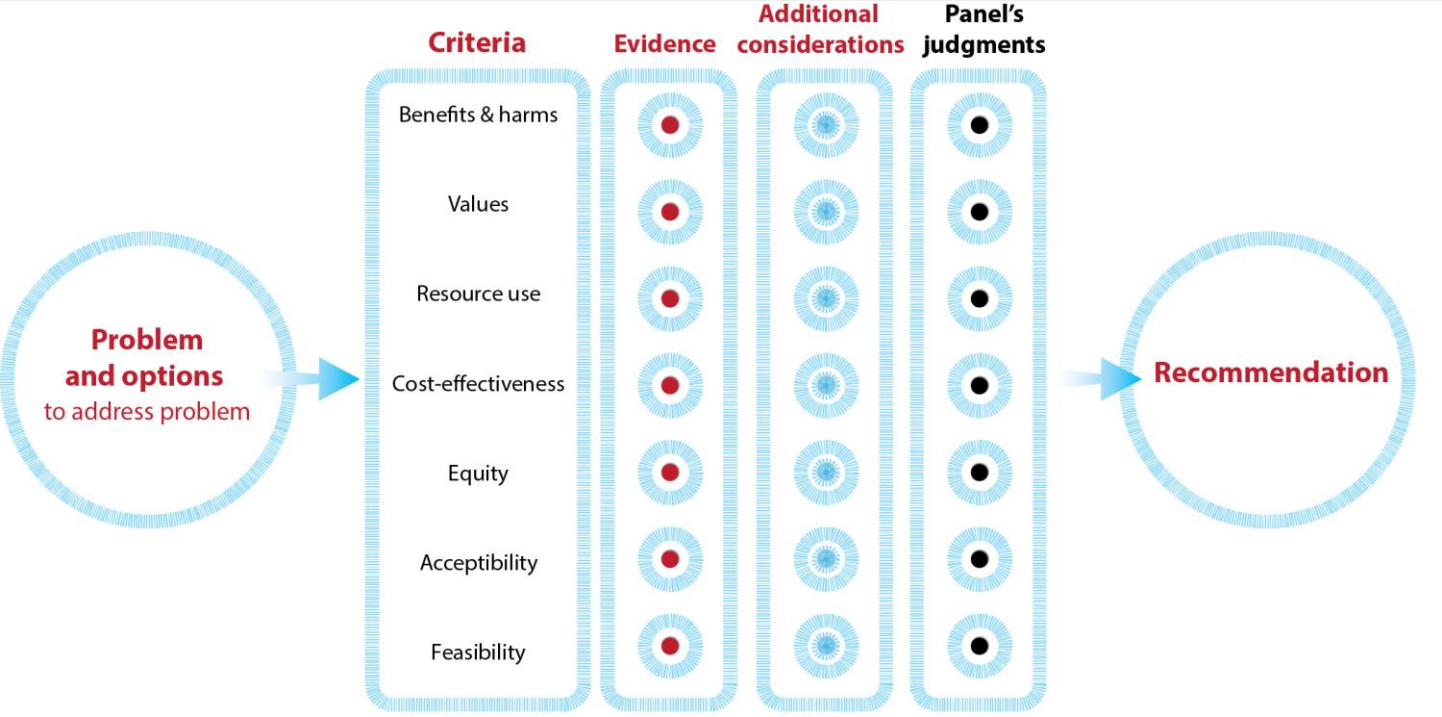


How recommendations are made

To a more systematic and transparent assessment of relevant criteria, and the evidence for these, in decision making...



Creating clarity about which judgements are made in relation to what information





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EVIDENCE-TO-DECISION COLLABORATION

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GRADE ETD Framework: Types of evidence

DOMAIN	Types of evidence
Problem a priority	
Balance of benefits versus harms	
Certainty of the evidence	
Values	
Resource requirements	
Cost-effectiveness	
Equity	
Acceptability	
Feasibility	

Guidance for using modelling for immunization decision-making. Geneva: World Health Organization; 2025. Licence: CC BY-NC-SA 3.0 IGO

GRADE ETD Framework: Types of evidence

DOMAIN	Judgement
Problem a priority	
Balance of benefits vs harms	
Certainty of the evidence	
Values	
Resource requirements	
Cost-effectiveness	
Equity	
Acceptability	
Feasibility	



Formulate question

Select outcomes

Rate importance

Outcomes across studies

Create evidence profile with GRADEPro

Rate quality of evidence for each outcome

P
I
C
O

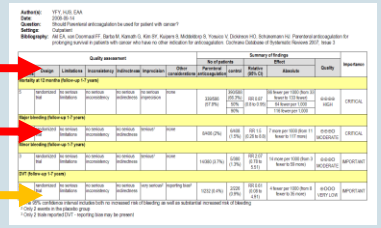
Outcome Critical

Outcome Critical

Outcome Important

Outcome Not important





Summary of findings & estimate of effect for each outcome

High
Moderate
Low
Very low

Grade down

1. Risk of bias
2. Inconsistency
3. Indirectness
4. Imprecision
5. Publication bias

Grade up

1. Large effect
2. Dose response
3. Opposing bias & Confounders

Systematic review

Guideline/HTA development

Decision-making panel

GRADE overall quality of evidence across outcomes based on lowest quality of *critical* outcomes

GRADE recommendations
- For or Against (Direction) ↓↑
- Strong or Conditional (Strength)

By considering balance of:



- ❑ Quality of evidence
- ❑ Balance of benefits/harms
- ❑ Values and preferences
- ❑ Resource utilization
- ❑ Equity and Human rights
- ❑ Acceptability/Feasibility

Guideline



AMERICAN GASTROENTEROLOGICAL ASSOCIATION

Colorectal Cancer Screening and Surveillance: Clinical Guidelines and Rationale—Update Based on New Evidence

Formulate Recommendations
- "The panel recommend using…"
- "The panel suggest using…"
- "The panel suggest not using…
- "The panel recommend not using…"



Evidence to Decision Frameworks

Provide

- Explicit and structured approach to decision making
- Transparency

- Does not prevent disagreement
  - Judgments are required
  - Experts in the field are in the best position to make these judgments
  - Disagreements need to be explored and built on

Making the recommendation

CONCLUSIONS

Type of recommendation

☐ Strong recommendation against the option	☐ Conditional recommendation against the option	☐ Conditional recommendation for either the option or the comparison	☒ Conditional recommendation for the option	☐ Strong recommendation for the option

Implications of a conditional recommendation

- Policy makers: There is a need for substantial debate and involvement of stakeholders
- Patients: The majority of people in this situation would want the recommended course of action, but many would not
- Clinicians: Be more prepared to help patients to make a decision that is consistent with their own values/decision aids and shared decision making

Implications of a strong recommendation

- Policy makers: The recommendation can be adapted as a policy in most situations
- Patients: Most people in this situation would want the recommended course of action and only a small proportion would not
- Clinicians: Most patients should receive the recommended course of action

Evidence is essential to inform decisions, but not sufficient – judgment is required

GRADE Evidence to Decision (EtD) frameworks

Can help guideline panels and decision makers move from evidence to a recommendation or decision by:

- Informing judgements about the pros and cons of each option (intervention)
- Considering each important factor that determines a decision (criteria)
- Providing a concise summary of the best available research evidence to inform judgements
- Helping to structure discussion and identify reasons for disagreements
- **Making the basis for decisions transparent and adaptable for target audiences**

Key take aways for you?

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Useful resources and readings

- World Health Organization. (2022). *Evidence, policy, impact: WHO guide for evidence-informed decision-making*. Geneva: World Health Organization. ISBN 978-92-4-003987-2.
- South African National Department of Health. 2025. *Evidence-Informed Decision-Making Manual for the National Standard Treatment Guidelines and the Essential Medicine List: Applying the National Standards for Evidence-Informed Decision-Making to Develop Trustworthy Guidance Products* [Draft]. Pretoria, South Africa
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- GRADE working group website: http://www.gradeworkinggroup.org/
- Grade software for systematic reviews and guidelines: https://gradepro.org/
- Guyatt GH, Oxman AD, Vist GE, Kunz R, Falck-Ytter Y, Alonso-Coello P, et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. BMJ. 2008;336(7650):924-6.
- Guidance for using modelling for immunization decision-making. Geneva: World Health Organization; 2025. Licence: CC BY-NC-SA 3.0 IGO content

Movement break